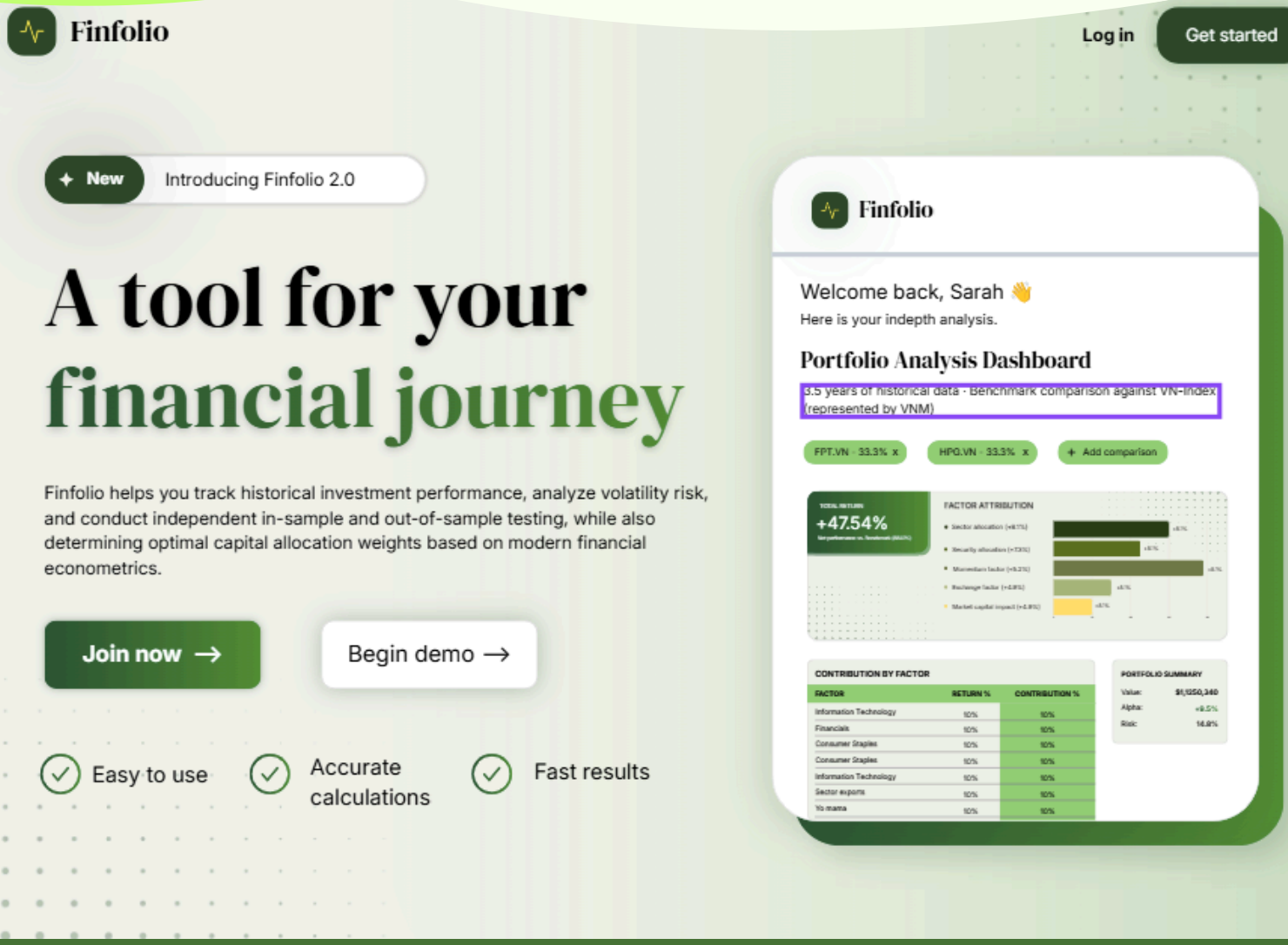
GROUP 9 PRESENTS

FINFOLIO 2.0



INDIVIDUAL REPORT

TRAN MINH NGOC

Table of Contents

INTRODUCTION AND INDIVIDUAL RESPONSIBILITY 3-4

DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0 5-8

IMPLEMENTATION AND DEVELOPMENT PROCESS 9

EVIDENCE, VERIFICATION AND PROJECT OUTCOMES 10

LIMITATIONS AND LESSONS LEARNED 11

CONCLUSION 12





Tran Minh Ngoc

Data Governance

- Input dictionary
- Source register
- Benchmark & Currency policy
- Classification & Freshness

I. INTRODUCTION AND INDIVIDUAL RESPONSIBILITY

1. The data problem

Finfolio 2.0 accepts portfolios that can contain Vietnamese equities and internationally traded commodity proxies. This scope creates a data problem before it creates a calculation problem. Instruments use different symbol conventions, prices are quoted in different currencies, and markets do not share identical holidays or closing times. A portfolio, benchmark and exchange-rate series can each look valid while remaining incompatible for a relative performance calculation. The project needed a documented contract connecting user input to provider observations and stating when a calculation may proceed.

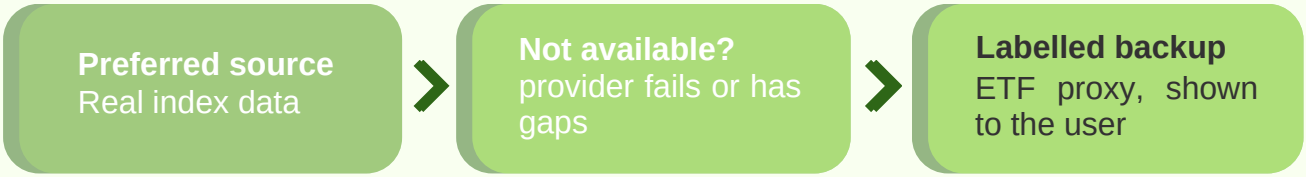
2. My role in this subject

I owned that contract: the input dictionary, source register, instrument metadata, benchmark roles, currency fields, alignment rules, data-quality checks and freshness language. I also specified sector and industry classification for instruments that do not represent operating companies.

I. INTRODUCTION AND INDIVIDUAL RESPONSIBILITY

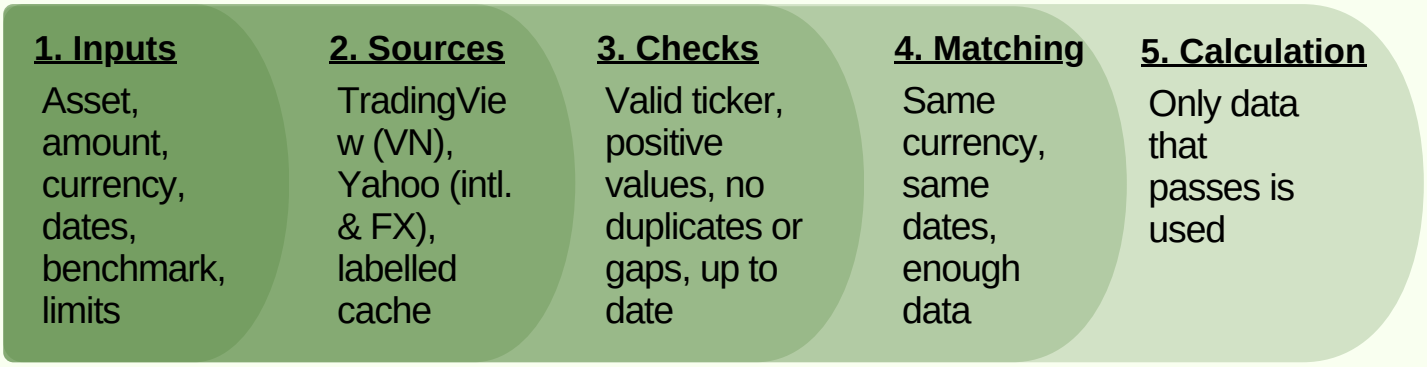
3. Personal highlight - the fallback must stay visible

- For every data source, I recorded three things: the preferred source, the backup if it fails, and the known limitations. You can find this in docs/WEEK3.md , the stock catalog files, the data provider code, and the tests.
- If the app quietly swaps in a backup, the meaning of the result changes without the user knowing. For example, an ETF that tracks VN30 is close to VN30, but not the same: it has fees and tracking differences. So whenever a backup is used, the app must label it clearly.



4. How data travels through the app

Every price goes through the same five steps before the app uses it in a calculation. At every step, we can still see the source, date, backup status and changes made.



> Every number used in a calculation should answer three questions: Where did it come from? In what currency? For which dates?

II. DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0

2.1. Input dictionary

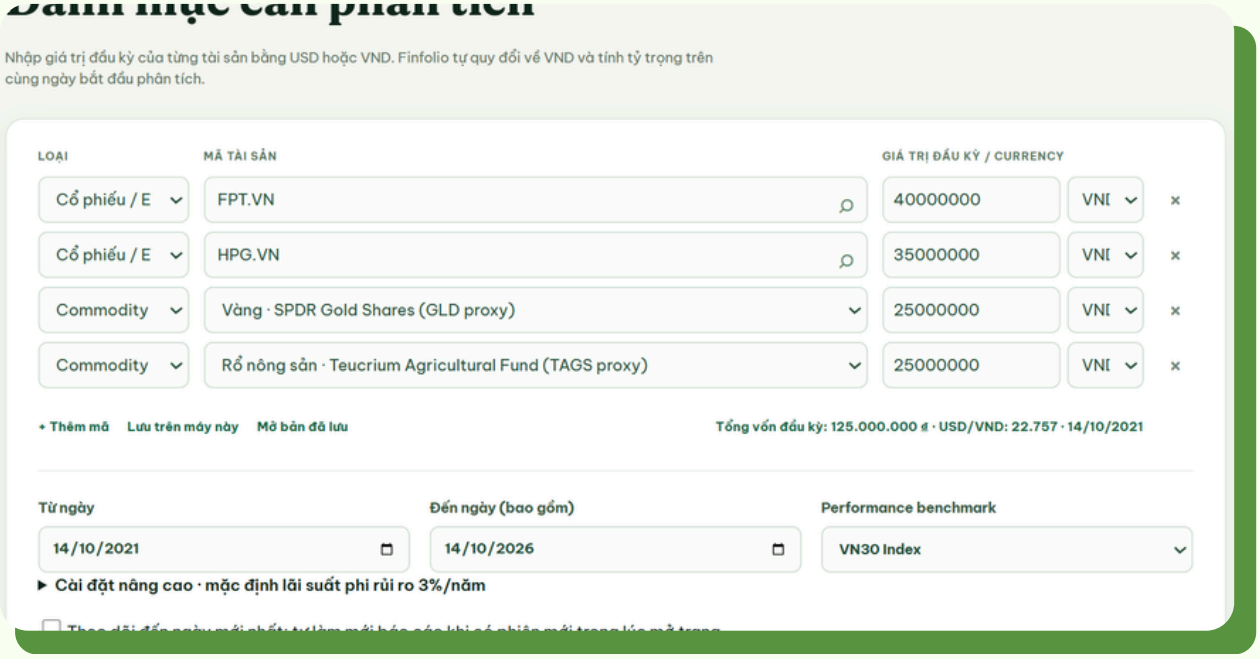
- The input dictionary is a list of every piece of information the app needs, written so that nobody can misread it. It covers the ticker, exchange, asset type, starting amount, currencies, sector and industry, price dates, benchmark, and allocation limits. For each field, it states the type, unit, an example, the validation rule, and who provides it.
- Design choice: users enter how much money they invested, not how many shares. Most users don't know a historical share count, so the app works that out itself.

Currency the user types in	≠	**Currency the asset trades in**
What the user chooses for the starting amount - VND or USD		Set by the market (e.g. FPT in VND, a gold ETF in USD)

Mixing these two up would give the wrong number of shares and the wrong return. So the app:

1. Converts the starting amount into the base currency on the start date;
2. Calculates the number of shares from the asset's own price;
3. Keeps the asset's trading currency to measure exchange-rate effects later.

The figure below shows the input form built on these rules. Each row has a separate asset type, ticker, amount and currency. The form also asks for the date range, benchmark and allocation limits up front.

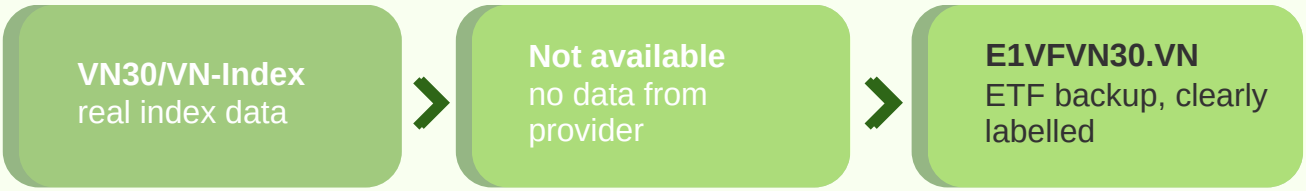


II. DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0

2.2. Data sources and the Vietnam benchmark

Source	Used for
TradingView (tvdatafeed)	Vietnamese stocks and indices. (Yahoo Finance could not give a stable VN Index history)
Yahoo Finance	International stocks, ETFs, commodity funds and exchange rates.
User input	Investment amounts and settings
Fixed test data	Automated tests only, always labelled as test data

For example, VN30 or VN-Index can be chosen as the benchmark, but they are never added to the portfolio. If the index data is unavailable, the app uses the E1VFVN30.VN ETF instead.



The label stays visible because an ETF has fees and does not track its index perfectly. Following feedback from outside reviewers, we also chose one clear benchmark rather than an automatic mix of several, because Finfolio's goal is simple performance comparison.

LIMITS OF FREE DATA SOURCES

Free providers may limit requests, change past data, or miss some details, so data as "downloaded today" does not mean "has today's prices". The app shows four things separately: the provider, when the data was downloaded, the latest market day it covers, and whether a backup was used.

II. DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0

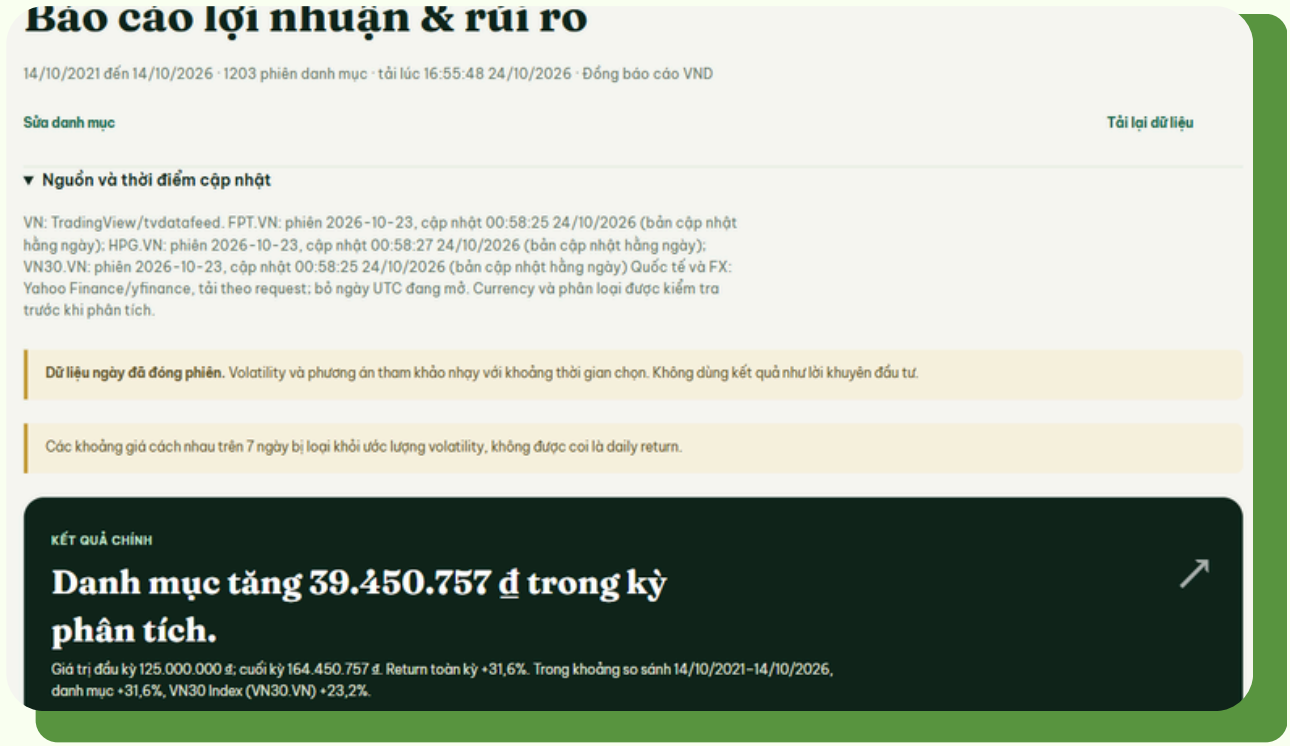
2.3. Matching dates and currencies

Vietnamese stocks, US assets, the benchmark and the USD/VND rate do not always trade on the same days. When the Vietnamese market has closed for the day, the US market may still be open. So the app only uses the days on which all the needed data exists.

> **RULE: DON'T COPY OLD PRICES ACROSS LONG HOLIDAYS**
> If we simply repeated the last price during a long market closure, the app would see fake “0% return” days. That would make risk look lower than it really is. So long gaps are left out of daily risk calculations.

Comparison measures (active return, tracking error, beta, alpha and information ratio) are only calculated when the portfolio and benchmark cover the same period with enough shared data. A VND portfolio is converted to VND before being compared with a VND benchmark.

- This way, a difference in dates or currency is never mistaken for good or bad investing.



2.4. Search and classification

The app's list of supported assets combines a Vietnam stock list and an international catalog. Each asset has:

Standard ticker	Full name	Market
Asset type	Trading currency	Sector / Industry

Users can search by ticker or company name, in upper or lower case, with or without Vietnamese accents. They don't need to know technical suffixes such as ".VN".

Problem: not every asset has a sector

"Sector" and "industry" make sense for companies, but not for gold, a broad ETF or crypto. Giving them a fake sector would be misleading. So I set a clear label: "Not applicable"; and kept commodity type (e.g. gold) as its own group.
As a result, users can break down their results in six ways, without any made-up labels:

Asset **Asset class** **Market/ commodity type**

Currency **Sector** **Industry**

III. IMPLEMENTATION AND DEVELOPMENT PROCESS

3.1. Work timeline

Weeks 1-2: Spot the risks

The app's list of supported assets combines a Vietnam stock list and an international catalog. Each asset has:

Week 3: Write the rules

Turned those risks into formal documents: input dictionary, source register, sample data, assumptions, data structure, validation rules and data flow. Everyone now knew what each field means and who provides it.

Weeks 4-5: Connect to formulas and screens

Linked benchmark and exchange-rate fields to the formulas that use them, and made sure search results show name, market and currency.

Weeks 6-7: Test it

Checked Vietnam data coverage, search, classification, backups and freshness with automated tests: *test_vn_data.py* , *test_global_data.py* , *freshness.test.mjs* .

3.2. Data freshness

In the final check, we found a data file that was downloaded recently but still contained old prices, so I changed how "fresh" is defined. The app now shows two dates:

Latest market day

The most recent trading day actually included in the data.

Last updated

When the system last downloaded the data.

A scheduled job checks the number of assets and the latest date before saving new data. If the data is missing or out of date, the server tries the provider again and shows an error instead of quietly using test data.

Limit: the schedule (GitHub Actions) may not run at the exact minute, and the app has no official holiday calendar. These limits are documented, and the app never invents prices for weekends or holidays

IV. EVIDENCE, VERIFICATION AND PROJECT OUTCOMES

4.1. Evidence of contribution

What I delivered	Where it is in the code	Verification procedure
Input dictionary & source	*docs/WEEK3.md*	*Trace every portfolio form field to type, unit, validation, and source.*
Vietnam assets & benchmarks	• *demo/vn_universe.json* • *demo/vn_data.py*	*Run the Vietnam data tests and inspect direct index or labelled proxy status.*
International assets	• *demo/instrument_catalog.jsondemo/global_data.py*	*Confirm symbol, market, quote currency, and instrument type.*
Search & classification	• *demo/classification.py* • *demo/symbol_search.py*	*Search by ticker and name; inspect non-corporate labels.*
Data freshness	• *demo/data_status.py* • *demo/freshness.mjs*	*Compare latest session with refresh timestamp and run the freshness tests.*

4.2. Contribution to final product

The final outcome was a data pipeline supporting an actual multi-asset example: Vietnamese equities and a USD commodity ETF proxy. The pipeline supplies:

Common VND basis	**Declared benchmark**	**Transparent source timing**

These conditions enabled the financial engine to calculate attribution and risk without hiding differences in provider, currency, or market calendar.

V. LIMITATIONS AND LESSONS LEARNED

5.1. Limitations

No point-in-time classification

Sector and industry histories are not yet maintained, so a company may be classified with current metadata when historical classification would be preferable.

Finite Vietnam universe

The supported Vietnam symbol list is limited.

No official holiday calendar

Exchange holidays are inferred from observations rather than a dedicated calendar, and markets close at different times.

Free data can change

Providers may change their response format or revise historical values.

5.2. Lessons learned

> A financial result is only as trustworthy as the data behind it.

We must always be able to trace a number's source, currency, date, changes, backups and checks. Just because a provider returns numbers does not mean they are ready to use for risk or benchmark analysis.
Writing these rules down also helped the team: everyone had the same definitions, and the QA owner had clear conditions to test against.

VI. CONCLUSION

My contribution established the data architecture and provenance controls of Finfolio 2.0. In summary, I:

- defined every input and its details;
- solved the Vietnam benchmark data problem;
- separated the user's currency from the asset's currency;
- set rules for matching dates across markets;
- created classification rules for all asset types;
- improved how the app shows whether data is up to date

> These outputs allowed the application to support multiple asset classes while retaining a transparent chain from user input to market observation and financial calculation.

INDIVIDUAL DECLARATION

I confirm that this report reflects the work that I am prepared to explain, calculate, demonstrate, or reproduce in Finfolio 2.0. Shared project artifacts are identified as collaborative where applicable, and my individual accountability is defined by the evidence presented in this report.

DECLARATION	
Full name	Tran Minh Ngoc
Student ID	2412380036
Signature	
Date	